



# Machine Learning Accelerators

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# Max

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- Computer Science B.Sc. (2020–2023)  
Friedrich Schiller University Jena
- Computer Science M.Sc. (2023–2025)  
Karlsruhe Institute of Technology
- Researcher (since 2024)  
Friedrich Schiller University Jena  
Scalable Analyses Lab
- Ph.D. (beginning 2026)  
Friedrich Schiller University Jena  
Scalable Analyses Lab





# Tamino

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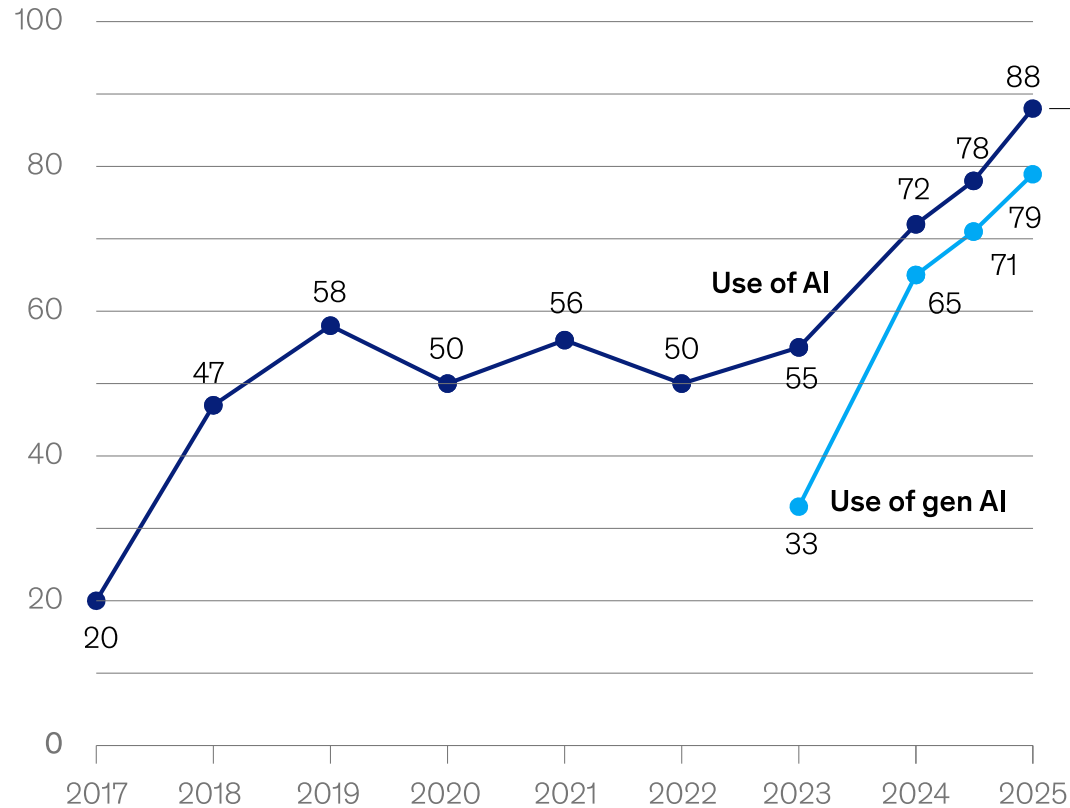
- Computer Science B.Sc. (2018–2022)  
Friedrich Schiller University Jena
- Computer Science M.Sc. (2022–2025)  
Friedrich Schiller University Jena
- Ph.D. (beginning 2026)  
Friedrich Schiller University Jena  
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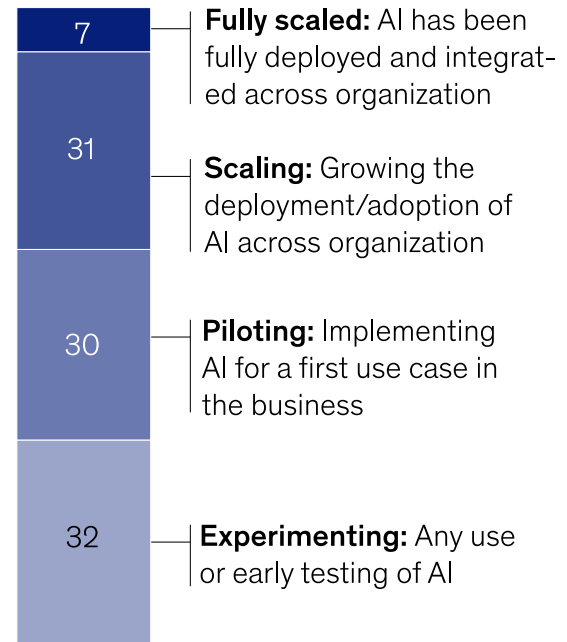
# Reported use of AI in at least one business function continues to increase.

Use of AI by respondents' organizations, % of respondents

Organizations that use AI in at least 1 business function<sup>1</sup>



Phase of AI use among organizations using AI in 2025



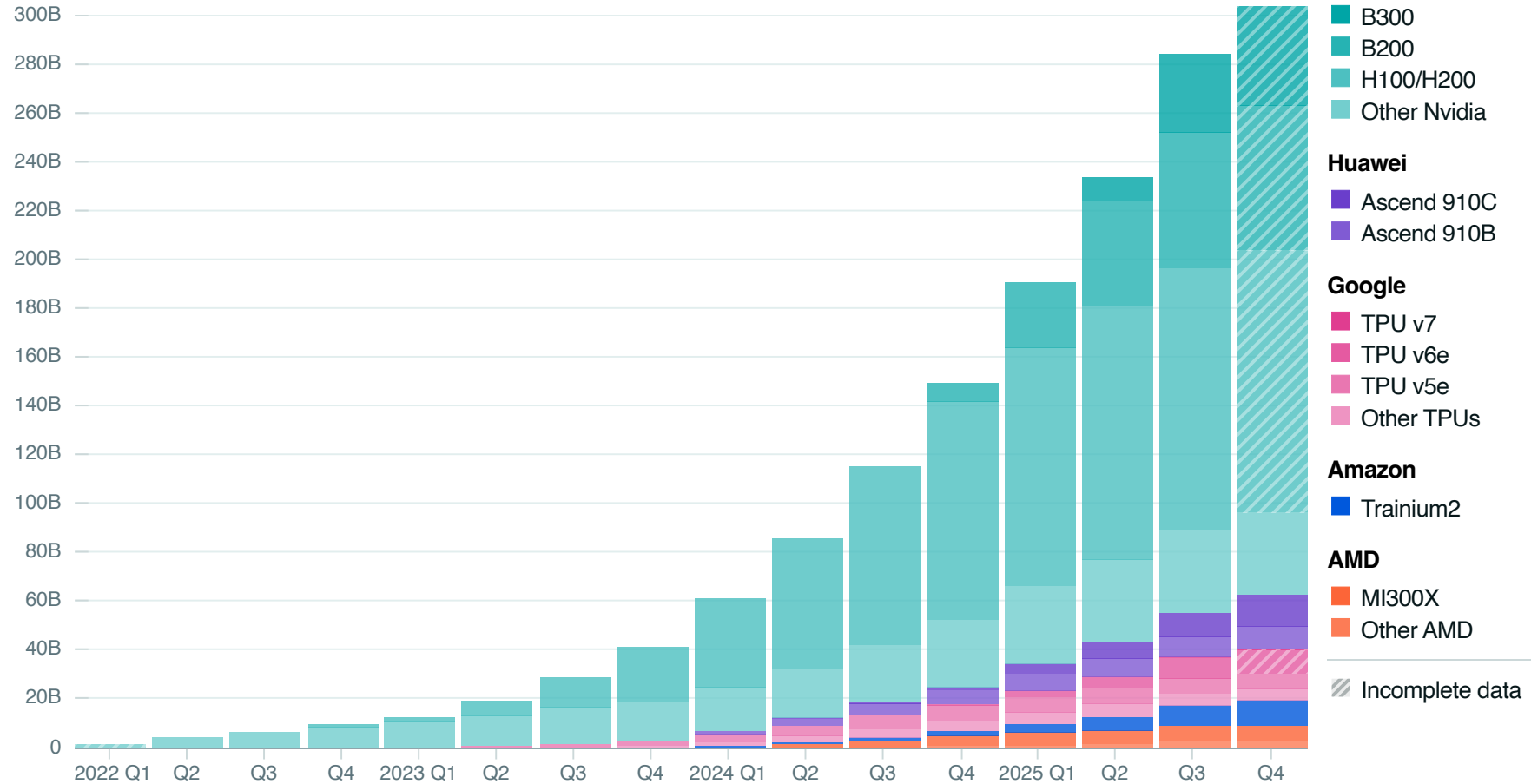
<sup>1</sup>In 2017, the definition for AI use was using AI in a core part of the organization's business or at scale. In 2018–19, the definition was embedding at least 1 AI capability in business processes or products. From 2020, the definition was that the organization has adopted AI in at least 1 function, and in 2025, the definition was regular use of AI in at least 1 function.

Source: McKinsey Global Surveys on the state of AI, 2017–25



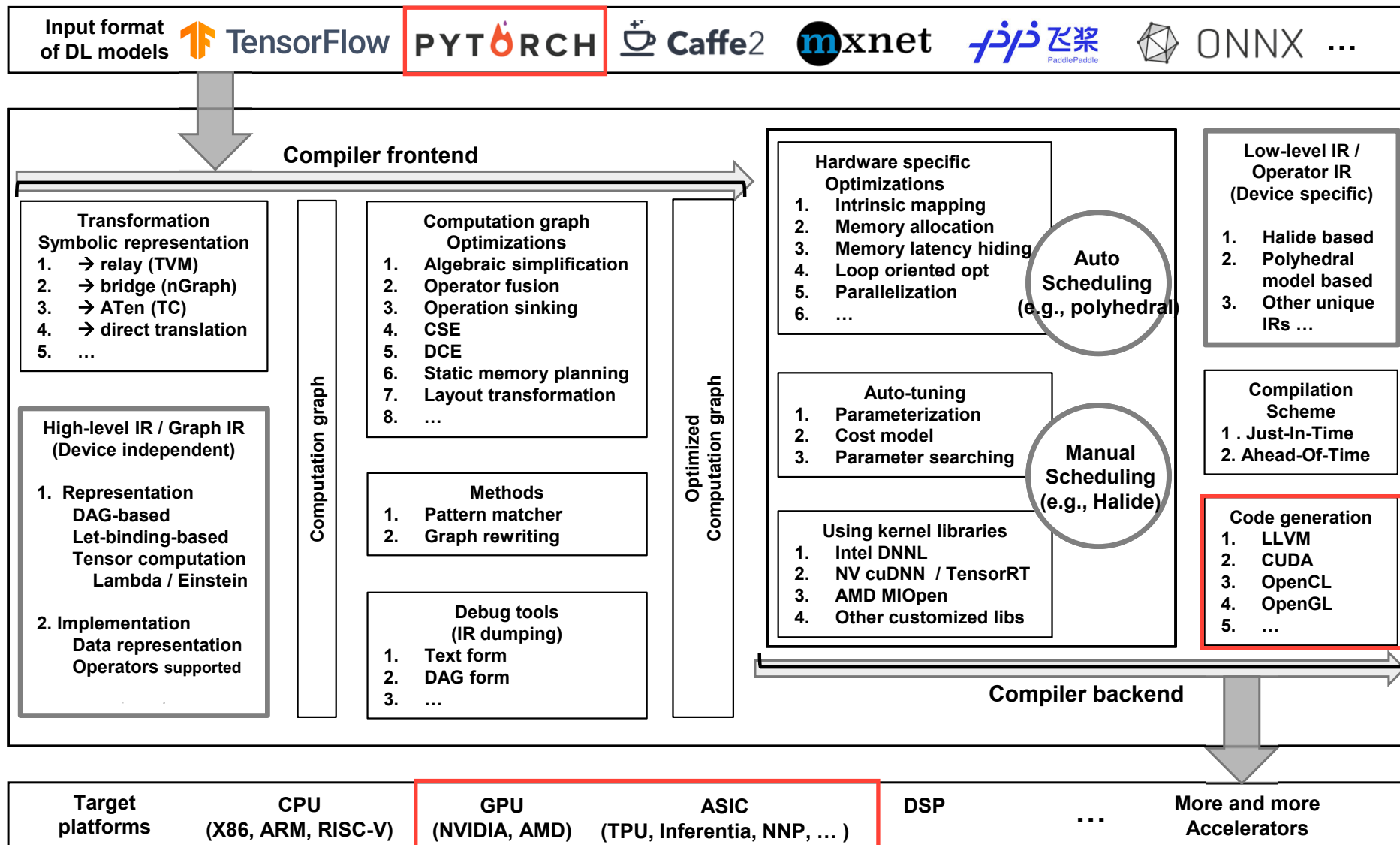
# AI Chip Sales

## Cumulative cost (USD)



Epoch AI, 'Data on AI Chip Sales'

Retrieved from: <https://epoch.ai/data/ai-chip-sales>, accessed 2026-02-13



# Roadmap

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- Background: PyTorch, Tensors, and Einsum
- GPU:
  - NVIDIA DGX Spark — GB10
  - cuTile
  - Tensor Cores
- NPU:
  - AMD Ryzen AI Max PRO 390 — XDNA2 NPU
  - Very Long Instruction Word ISA
  - Spatial Data Flow
- Project Implementation

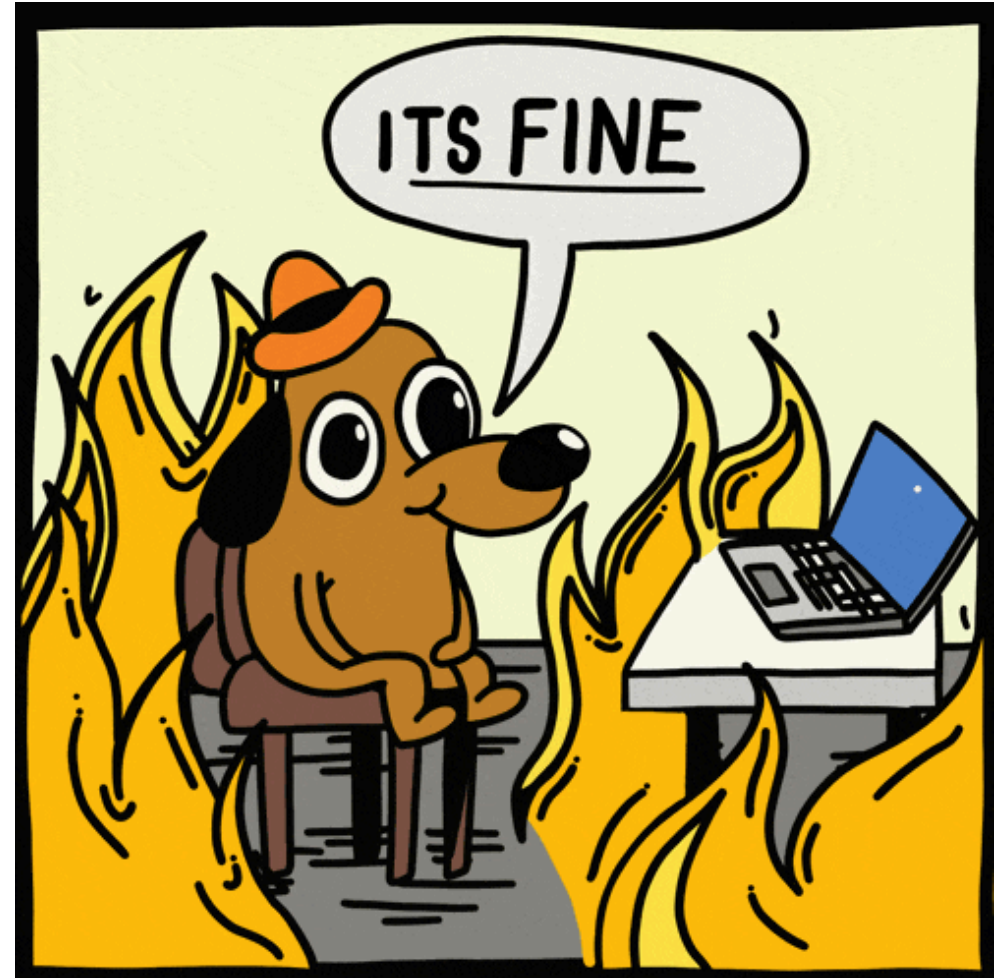




# This Course is...

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- First of a kind!
  - Everything is new!
  - Reflects the state of the art and beyond!
  - Might have a rough edges here and there 😊
- Interactive!
  - Hands-on experiments are a first class citizen!
  - Apply everything you learned!
  - Ask questions!



<https://c.tenor.com/OCAVH-SFsM8AAAAAd/tenor.gif>

# Prerequisites

- Good knowledge of computer architecture and low-level programming, e.g., assembly.
- Comfortable working with C/C++ and Python.
- Good knowledge of working with a Linux distribution.
- Knowledge of parallel programming (shared memory).
- Good knowledge of linear algebra.

```
1 bfp16_emu_mac:                                     // @bfp16_emu_mac
2 // %bb.0:                                           // %entry
3   nopa; vldb x2, [p1, #0]; nopxm; nops
4   vldb x4, [p1, #64]
5   nop
6   nop
7   nop
8   movxm r0, #16256
9   vlda.conv.fp32.bfp16 cml0, [p0, #0]; vbcst.16 x0, r0
10  mova r0, #53; vmov x1, x0
11  mova r1, #52; vshuffle x7, x2, x4, r0
12  vlda.conv.fp32.bfp16 cmh0, [p0, #64]; vshuffle x6, x2, x4, r1
13  mova r0, #60
14  vmul.f dm1, y3, y0, r0
15  nop
16  nop
17  vlda bml0, [p2, #0]
18  vlda bmlh0, [p2, #64]
19  vlda bmhl0, [p2, #128]; vconv.bfp16ebs8.fp32 ex0, dm0
20  vlda bmhh0, [p2, #192]; vconv.bfp16ebs8.fp32 ex2, dm1
21  nop
22  nop
23  mova r0, #780
24  vmac.f dm0, dm0, ex0, ex2, r0
25  nop
26  nop
27  nop
28  nop
29  ret lr
30  vst bml0, [p2, #0] // Delay Slot 5
31  vst bmlh0, [p2, #64] // Delay Slot 4
32  vst bmhl0, [p2, #128] // Delay Slot 3
33  vst bmhh0, [p2, #192] // Delay Slot 2
34  nop // Delay Slot 1
35 .Lfunc_end0:
36 .size bfp16_emu_mac, .Lfunc_end0-bfp16_emu_mac
```

# Organization



# Matrix Room

- Room on the Uni Jena Matrix Server called #mla:uni-jena.de
- Access using your university login data (or any other Matrix account)
- Course announcements will only be made in the room. Communicate with us via Matrix, that is do not write us e-mails!
- Use common sense when navigating the virtual space; for example:
  - You may use the room to ask questions.
  - You may use the room to discuss course-related topics with your peers.
  - Before the due dates, you may share solutions only with your direct teammates.

## General

Room Name  
Machine Learning Accelerators

Room Topic  
<https://github.com/scalable-analyses/mla>

Cancel

Save

## Room Addresses

### Published Addresses

Published addresses can be used by anyone on any server to join your room. To publish an address, it needs to be set as a local address first.

Main address  
#mla:uni-jena.de

☐ Publish this room to the public in uni-jena.de's room directory?



# Prüfungsleistung und Prüfungszulassung

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- Format:
  - Projekt im Verlauf des Semesters
  - Arbeit in Zweiergruppen ist erlaubt
  - 10x “reguläre” Abgaben mit je 0–10 Punkten
  - Gruppenspezifische Projektkomponente
- Prüfungszulassung:
  - mind. 50 Punkte bei den 10 “regulären” Abgaben
  - mind. 2x Präsentationen der “regulären” Abgaben; jedes Gruppenmitglied muss präsentieren
  - Rechtzeitig zur Prüfung anmelden!
- Prüfungsleistung:
  - Abschlusspräsentation (30%) am **08.07.2026**
  - Finale Abgabe (70%):
    - Projektbericht
    - Implementierung
    - **Abgabe bis 27.07.2026**
  - Falls Krankheit die Projektarbeit beeinträchtigt, bitte sofort Bescheid geben. Wir finden dann eine Lösung.

# Software Engineering and Submissions

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- Code and sources of project report have to be hosted in a git repository
  - Public or
  - Private — make the repository accessible to us
- Document your source code
- Write meaningful commit messages
- Take “issues” seriously, e.g., compiler warnings, and do not take shortcuts: no hacks!
- Test your software and your user guides on the reference machine(s)
- Use Sphinx to write your project report
- Assignments:  
<https://github.com/scalable-analyses/mla>
- Use Moodle for submissions
- Every team member has to submit! No submission means 0 points, no exceptions!
- Upload a single tar.xz file to Moodle which contains:
  - Link to git repository in `git_link.txt`
  - Git tag of the submission in `git_tag.txt`
  - Link to GitHub Pages in `report_link.txt` or current project report as HTML in `report/`
  - Briefly address the contribution of every team member in `contribution.txt`
- The deadline for regular submissions is Monday at 9 AM

- No cheating!
- Use of GenAI is allowed. You must understand and be able to present the results!
- Keep the project report to the point!
- Do **NOT** share your results before the submission deadline!
- Do **NOT** copy the results of others!



## Good Research Practice

Information on statutes, ombudspersons and seminars to ensure good research practice at the Friedrich Schiller University Jena

Women in the lab

Image: Jan-Peter Kasper (University of Jena)